



Diffusion Models

<https://hku-data8018.github.io/>

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Spring 2026



- Slides reference
 - MIT EECS Fall 2024, 6.S978 Deep Generative Models, <https://mit-6s978.github.io/>
 - <https://lilianweng.github.io/posts/2021-07-11-diffusion-models/>
 - Stanford Fall 2023, CS236 Deep Generative Models, <https://deepgenerativemodels.github.io/>

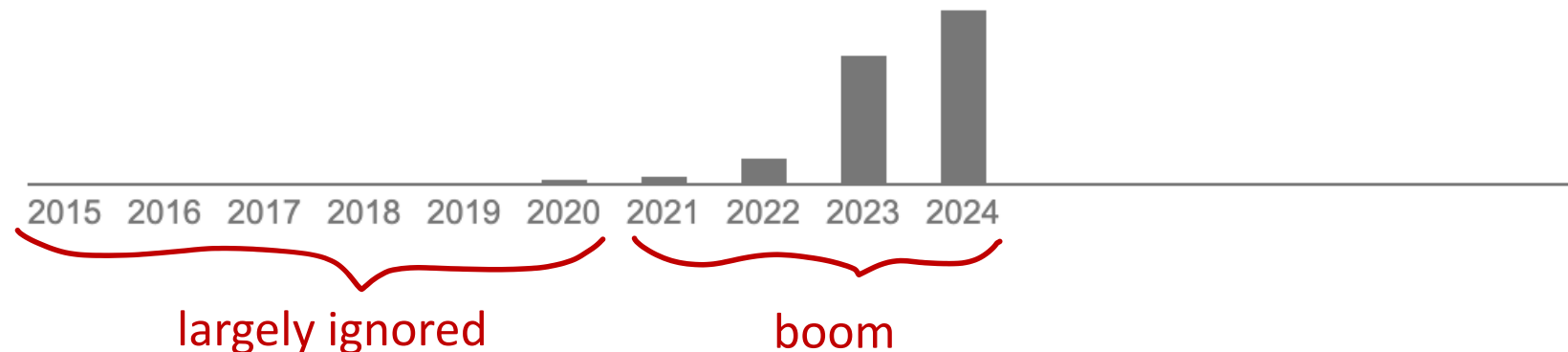
Deep unsupervised learning using nonequilibrium thermodynamics

Authors Jascha Sohl-Dickstein, Eric A Weiss, Niru Maheswaranathan, Surya Ganguli

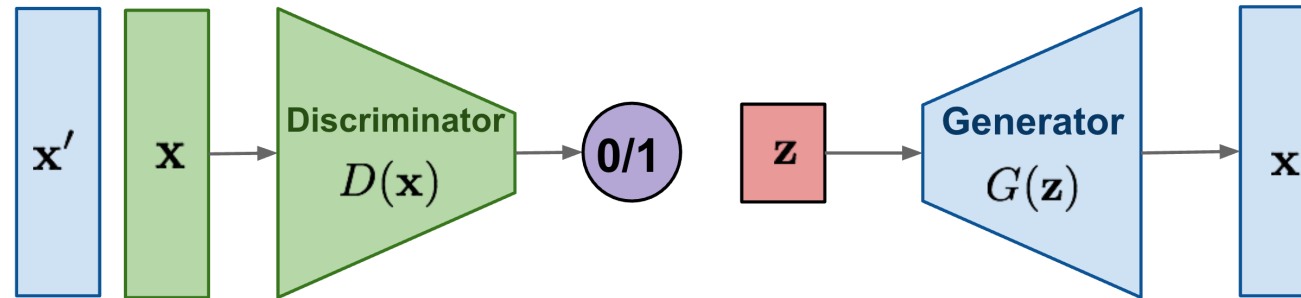
Publication date 2015/3/12

Journal International Conference on Machine Learning

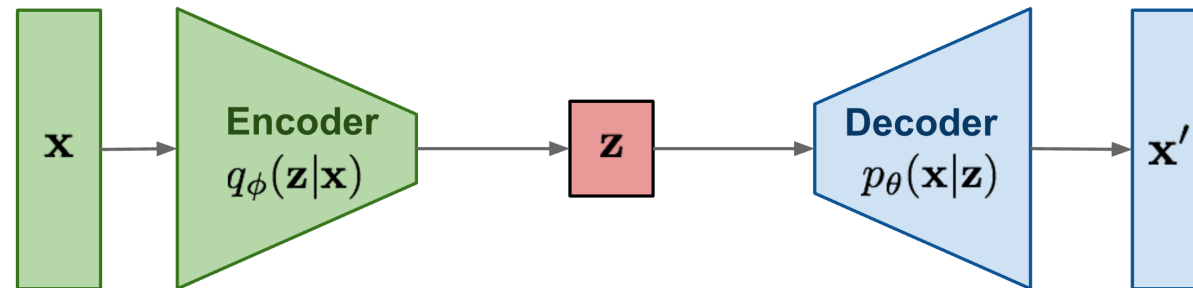
Total citations Cited by 5630



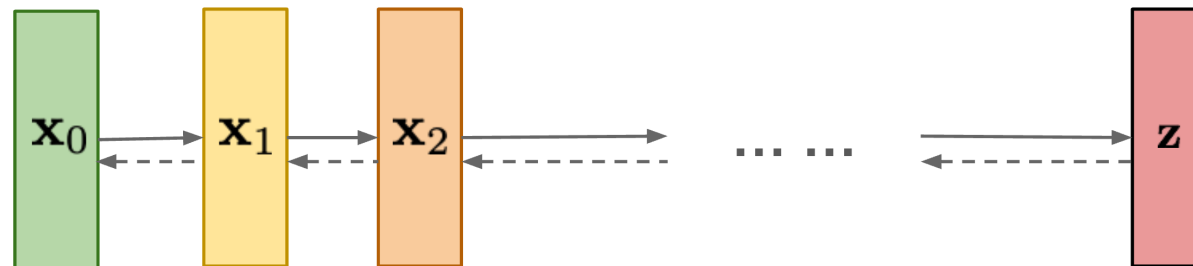
GAN: Adversarial training

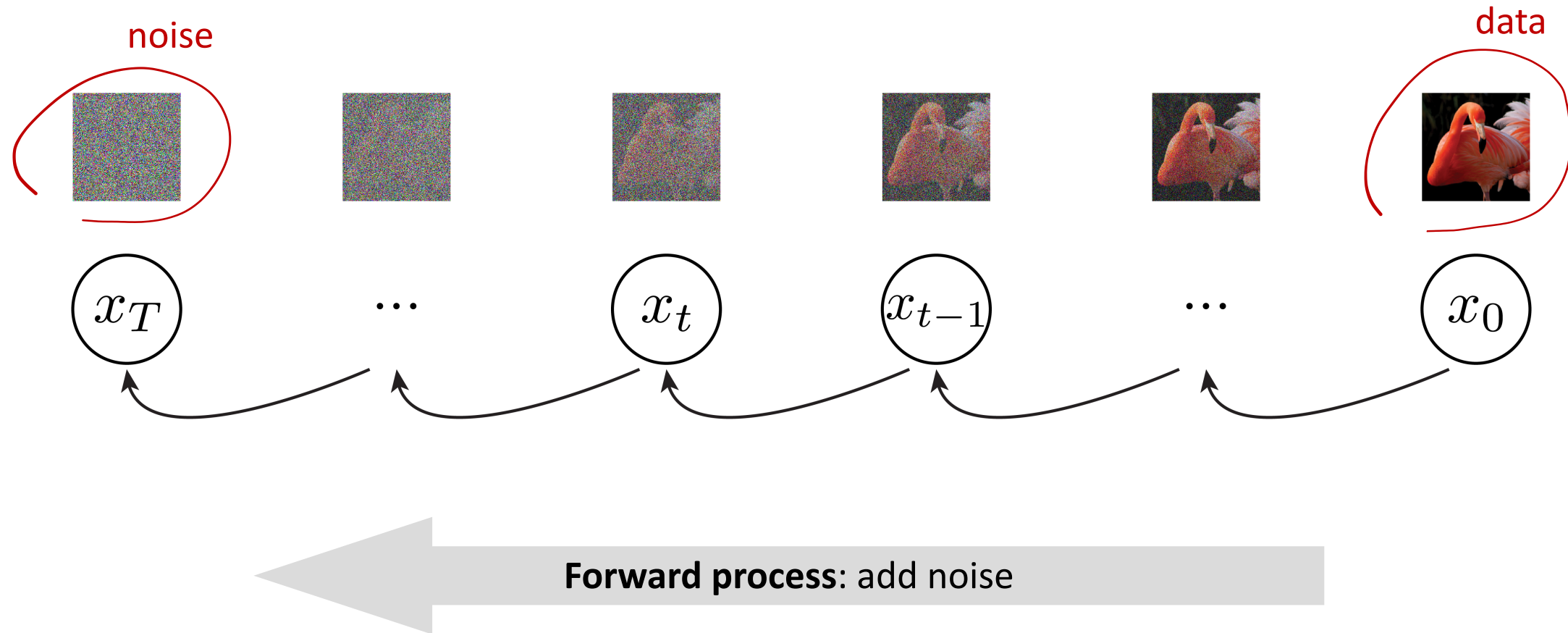


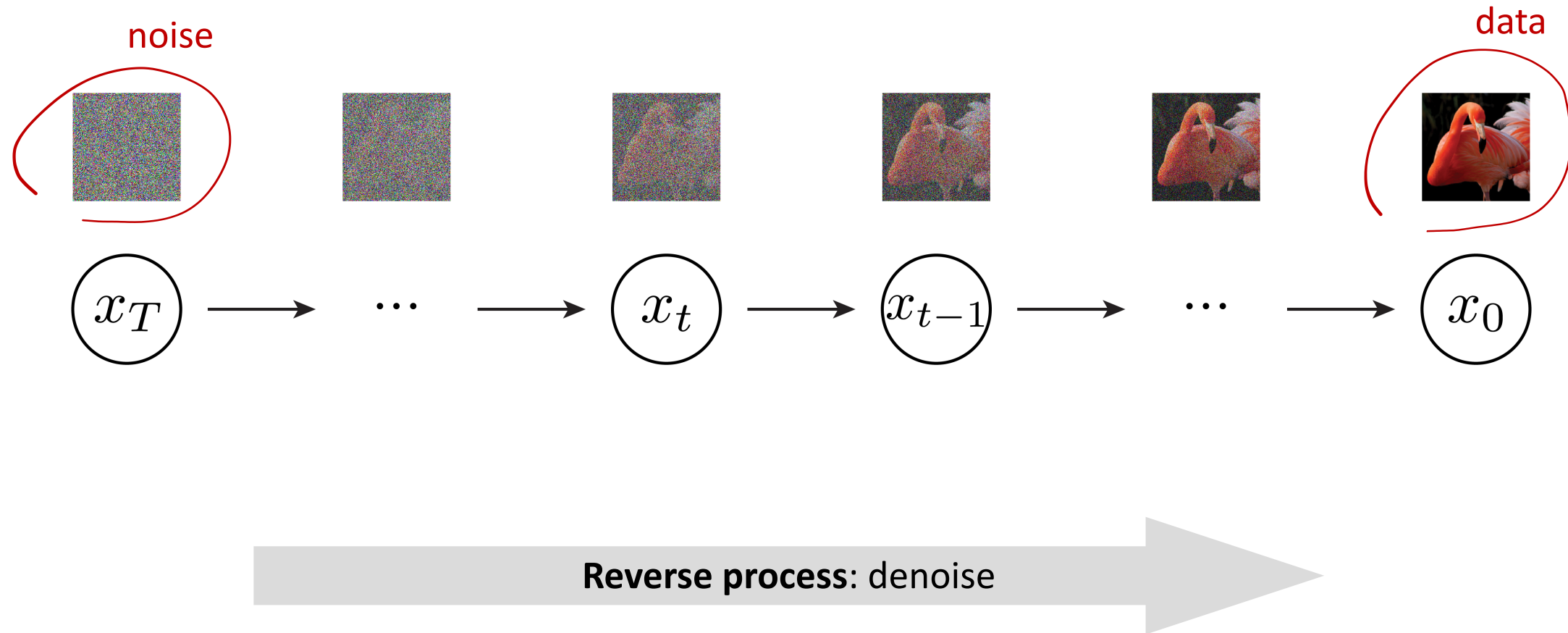
VAE: maximize variational lower bound



Diffusion models:
 Gradually add Gaussian noise and then reverse

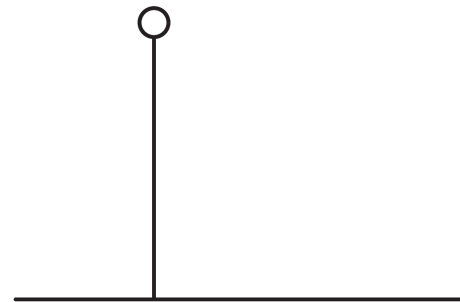




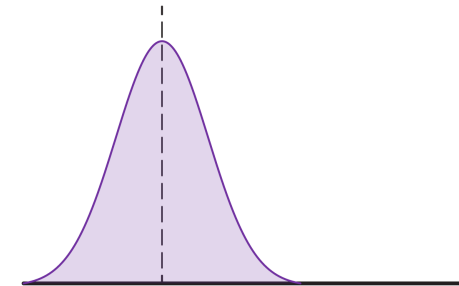


What is noise?

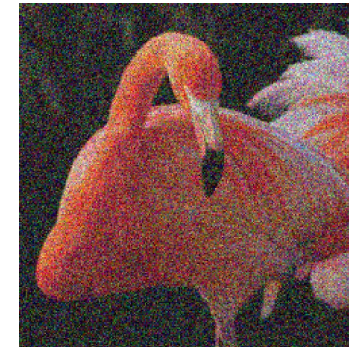
Adding Gaussian noise $\Leftrightarrow$ sampling $x \sim \mathcal{N}(x \mid x_0, \sigma)$



$$p(x) = \delta(x - x_0)$$

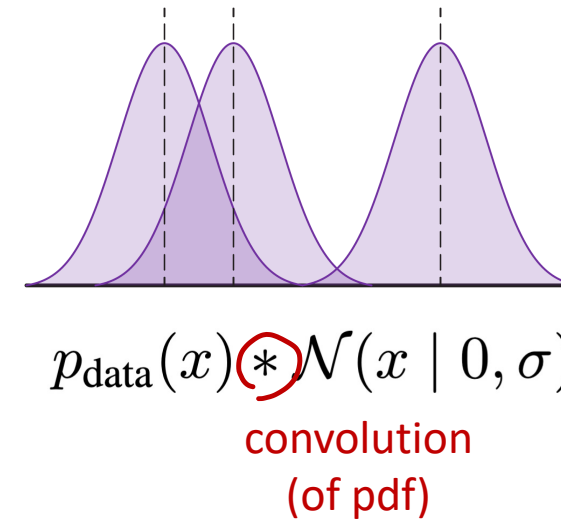
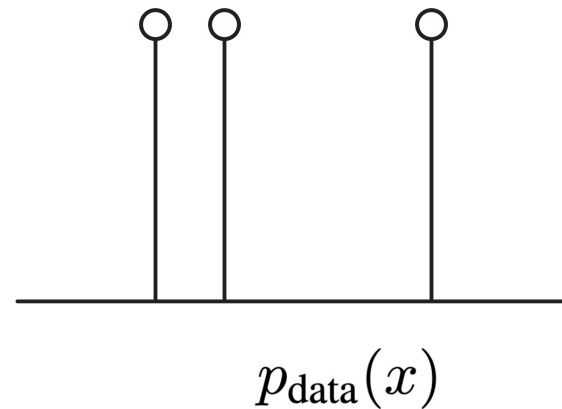


$$p(x) = \mathcal{N}(x \mid x_0, \sigma)$$



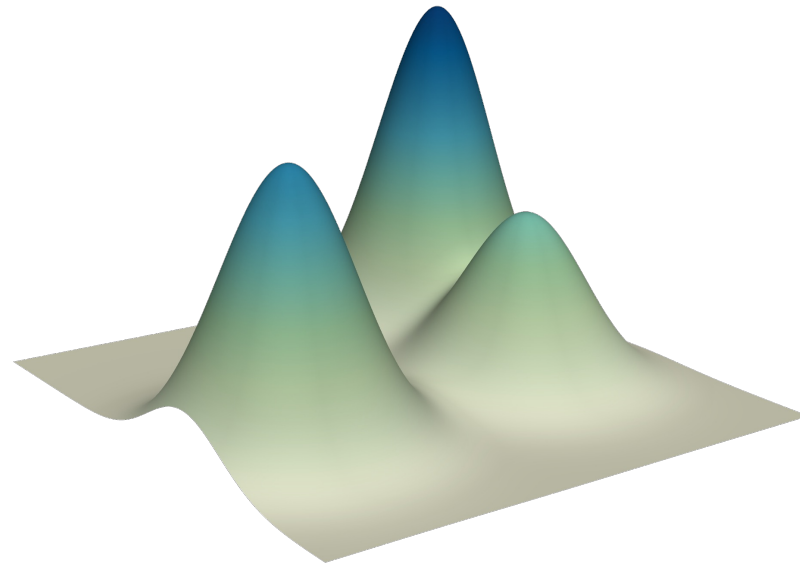
What is noise?

Adding Gaussian noise $\Leftrightarrow$ sampling $x \sim \mathcal{N}(x \mid x_0, \sigma)$

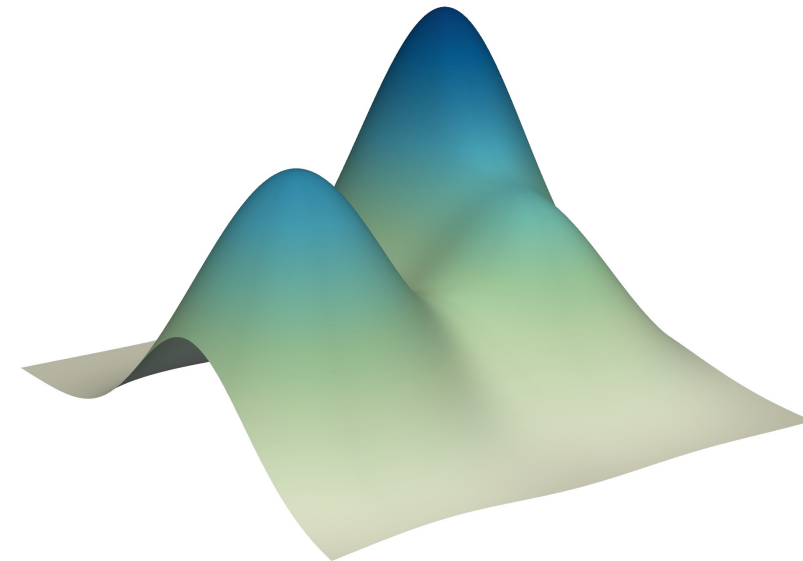


What is noise?

Adding Gaussian noise $\Leftrightarrow$ sampling $x \sim \mathcal{N}(x \mid x_0, \sigma)$



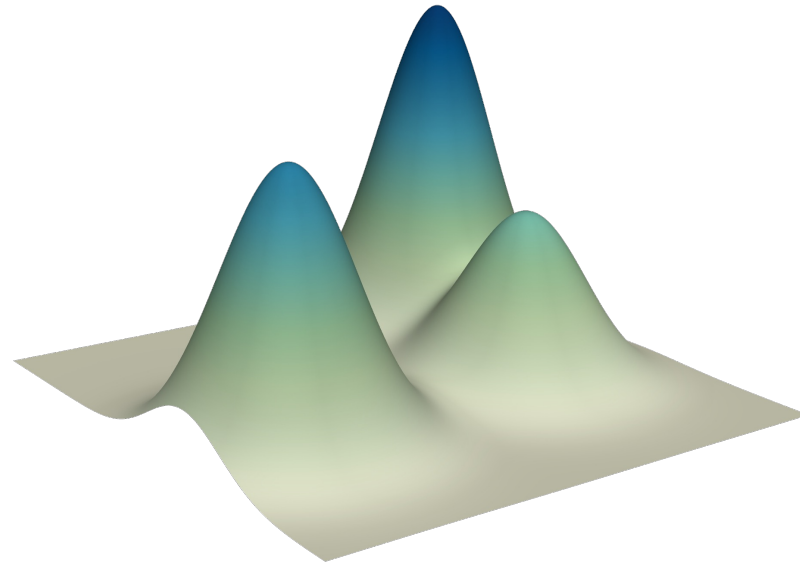
$p_{\text{data}}(x)$



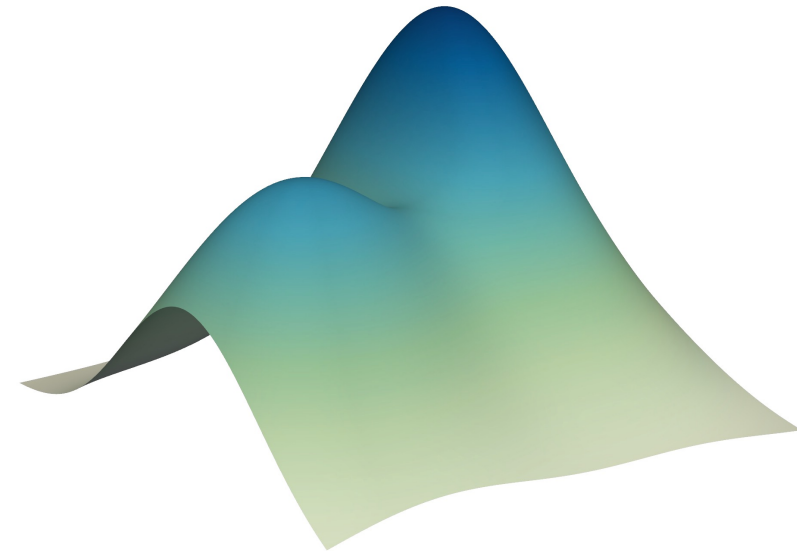
$p_{\text{data}}(x) * \mathcal{N}(x \mid 0, \sigma)$

What is noise?

Adding Gaussian noise $\Leftrightarrow$ sampling $x \sim \mathcal{N}(x \mid x_0, \sigma)$



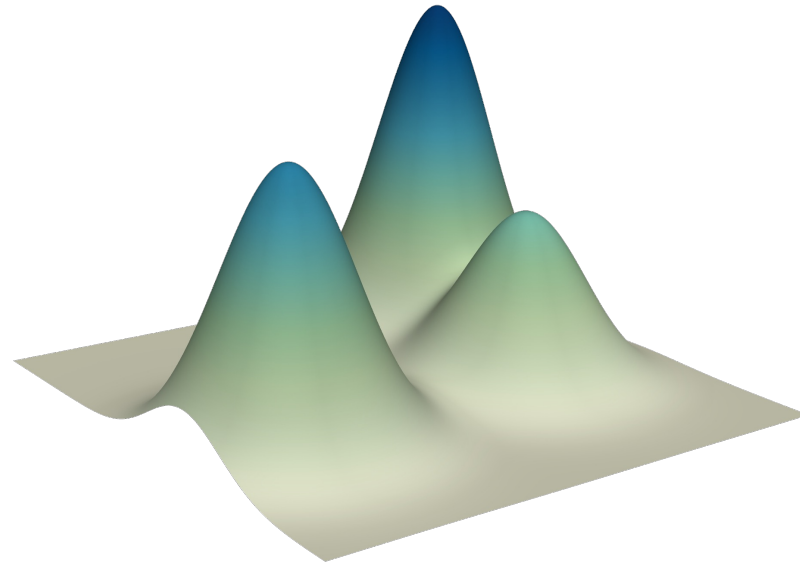
$p_{\text{data}}(x)$



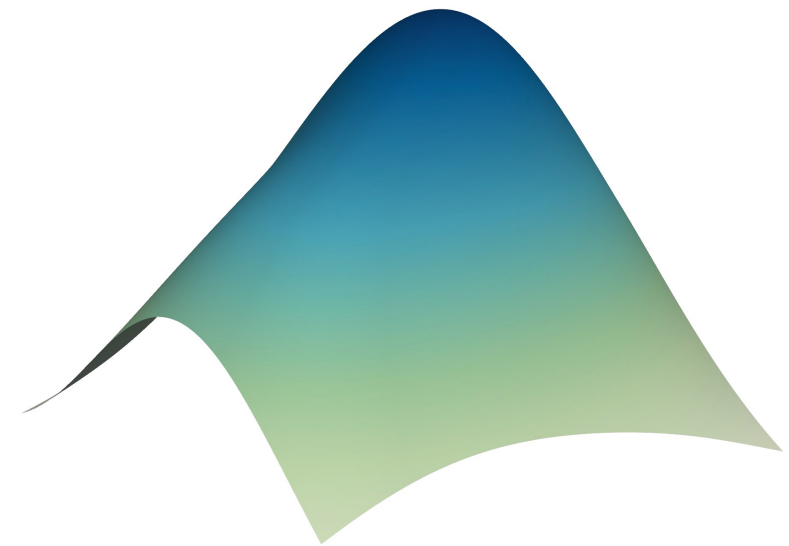
$p_{\text{data}}(x) * \mathcal{N}(x \mid 0, \sigma)$

What is noise?

Adding Gaussian noise $\Leftrightarrow$ sampling $x \sim \mathcal{N}(x \mid x_0, \sigma)$



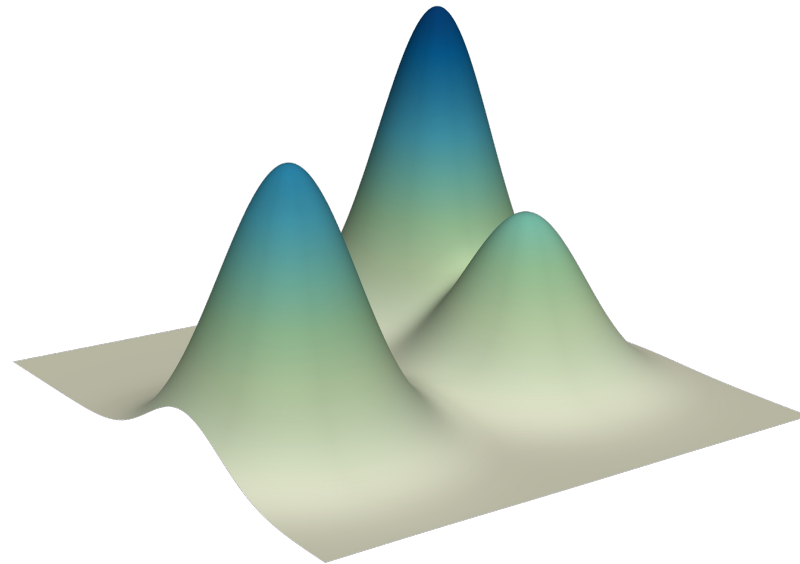
$p_{\text{data}}(x)$



$p_{\text{data}}(x) * \mathcal{N}(x \mid 0, \sigma)$

What is noise?

Adding Gaussian noise $\Leftrightarrow$ sampling $x \sim \mathcal{N}(x \mid x_0, \sigma)$

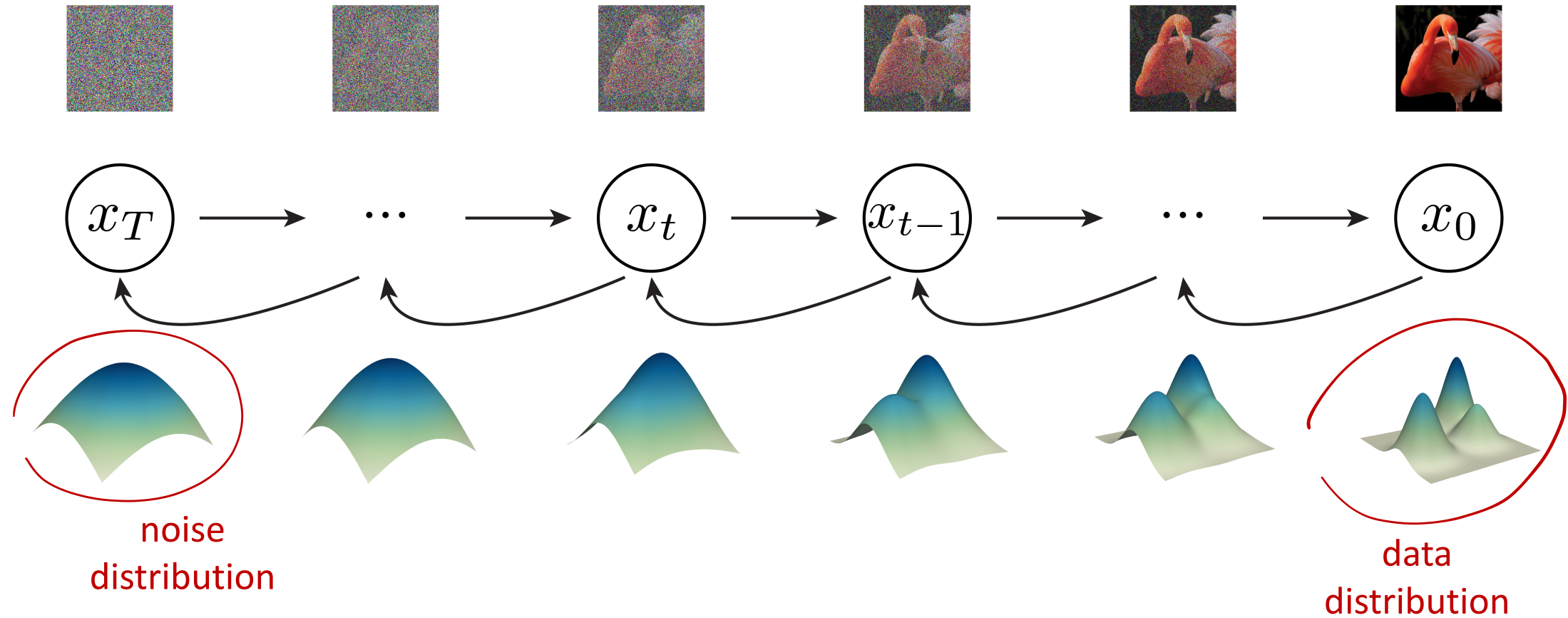


$p_{\text{data}}(x)$

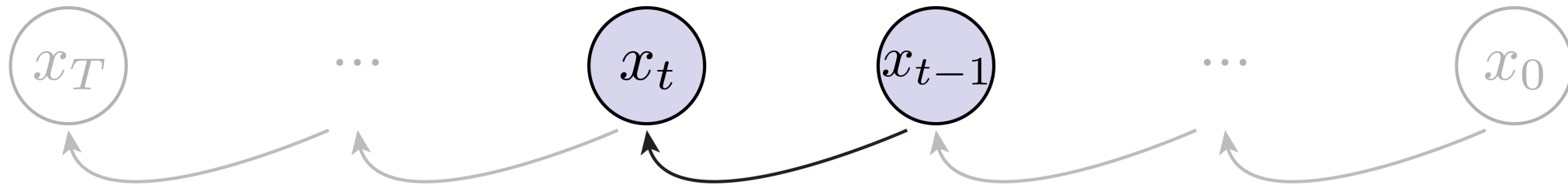


$p_{\text{data}}(x) * \mathcal{N}(x \mid 0, \sigma)$

What is noise?



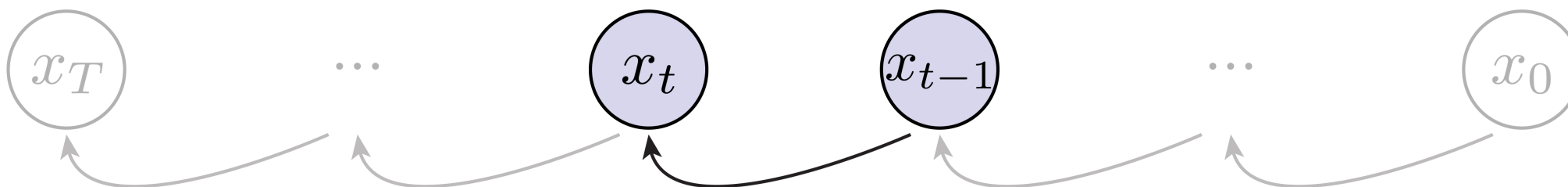
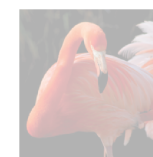
Forward Process



$$x_t = \sqrt{1 - \beta_t} x_{t-1} + \sqrt{\beta_t} \epsilon, \quad \epsilon \sim \mathcal{N}(0, \mathbf{I})$$

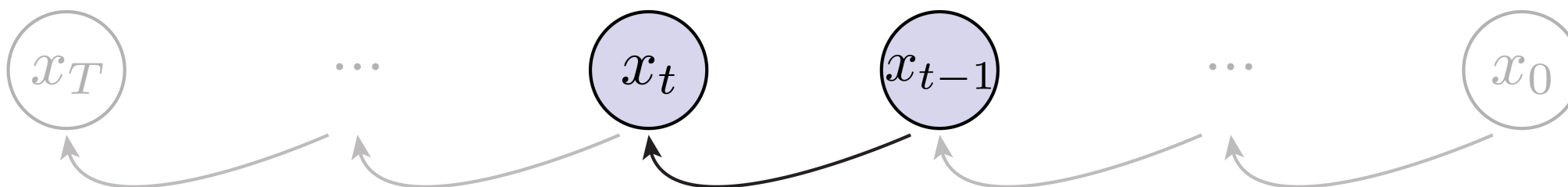
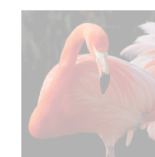
coefficients:
variance preserving

Forward Process



$$x_t = \underbrace{\sqrt{1 - \beta_t}}_{\text{mean of } x_t} x_{t-1} + \underbrace{\sqrt{\beta_t}}_{\text{std of } x_t} \epsilon, \quad \epsilon \sim \mathcal{N}(0, \mathbf{I})$$

Forward Process

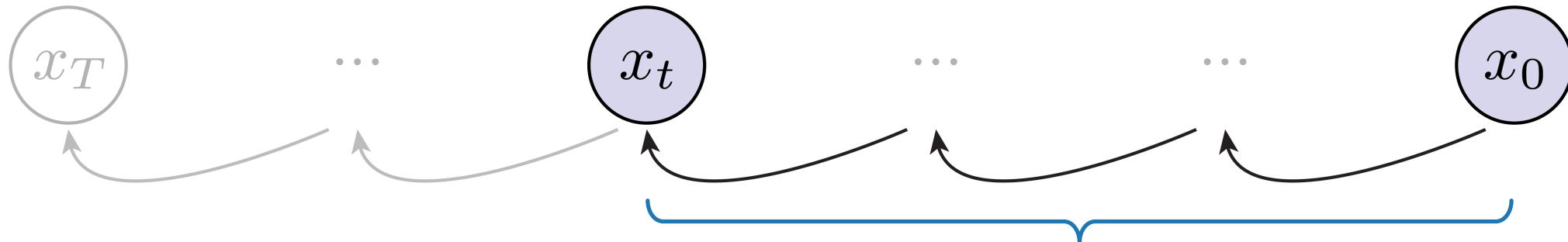


$$q(x_t \mid x_{t-1})$$

$$= \mathcal{N}(x_t \mid \underbrace{\sqrt{1 - \beta_t} x_{t-1}}_{\text{mean of } x_t}, \underbrace{\beta_t \mathbf{I}}_{\text{var of } x_t})$$

Forward Process

- Usually **a larger update step when the sample gets noisier**, so $\beta_1 < \beta_2 < \dots < \beta_T$ and $\bar{\alpha}_1 > \dots > \bar{\alpha}_T$



Let $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{i=1}^t \alpha_i$:

$$\mathbf{x}_t = \sqrt{\alpha_t} \mathbf{x}_{t-1} + \sqrt{1 - \alpha_t} \boldsymbol{\epsilon}_{t-1}$$

$$= \sqrt{\alpha_t \alpha_{t-1}} \mathbf{x}_{t-2} + \sqrt{1 - \alpha_t \alpha_{t-1}} \bar{\boldsymbol{\epsilon}}_{t-2}$$

$= \dots$

$$= \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}$$

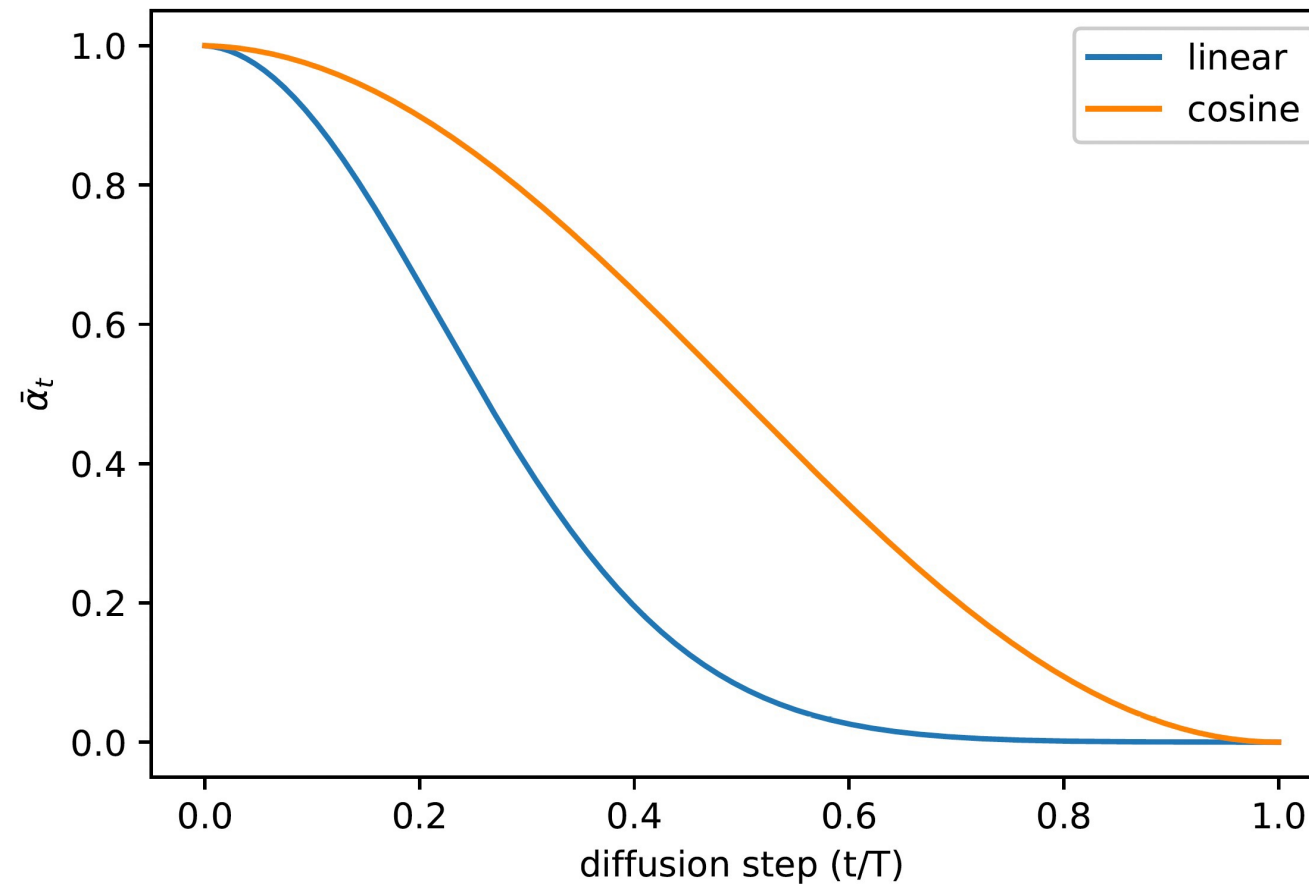
$$q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I})$$

; where $\boldsymbol{\epsilon}_{t-1}, \boldsymbol{\epsilon}_{t-2}, \dots \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$

; where $\bar{\boldsymbol{\epsilon}}_{t-2}$ merges two Gaussians (*).

Forward Process

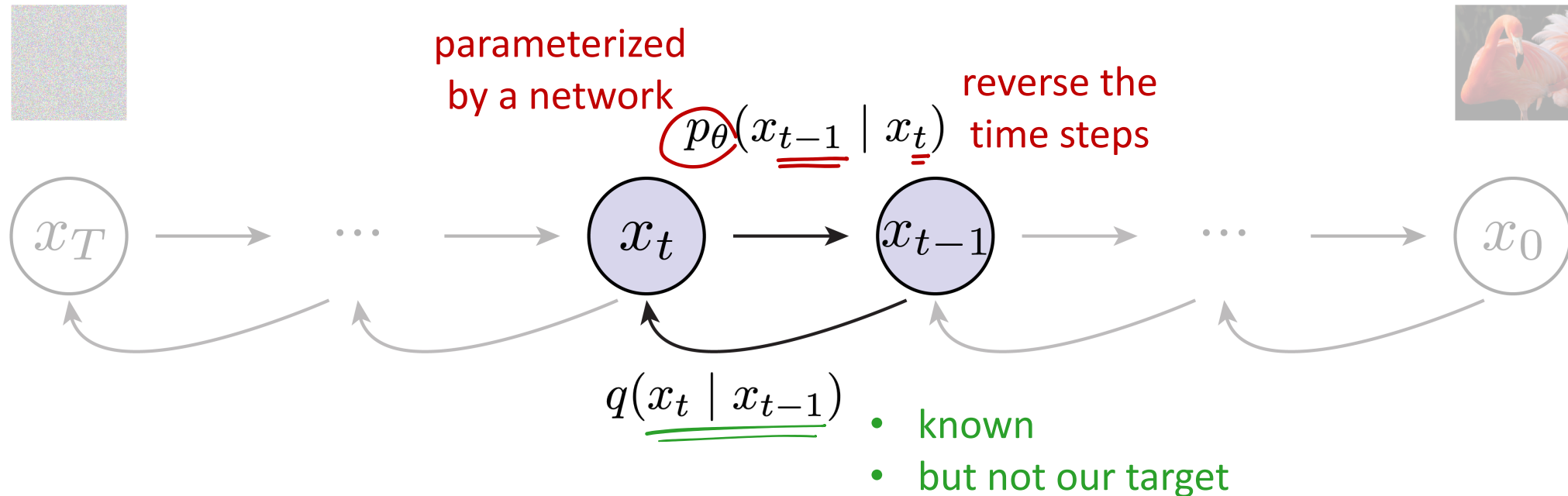
- Usually a larger update step when the sample gets noisier, so $\beta_1 < \beta_2 < \dots < \beta_T$ and $\bar{\alpha}_1 > \dots > \bar{\alpha}_T$
- **Noise scheduler**



Reverse Process

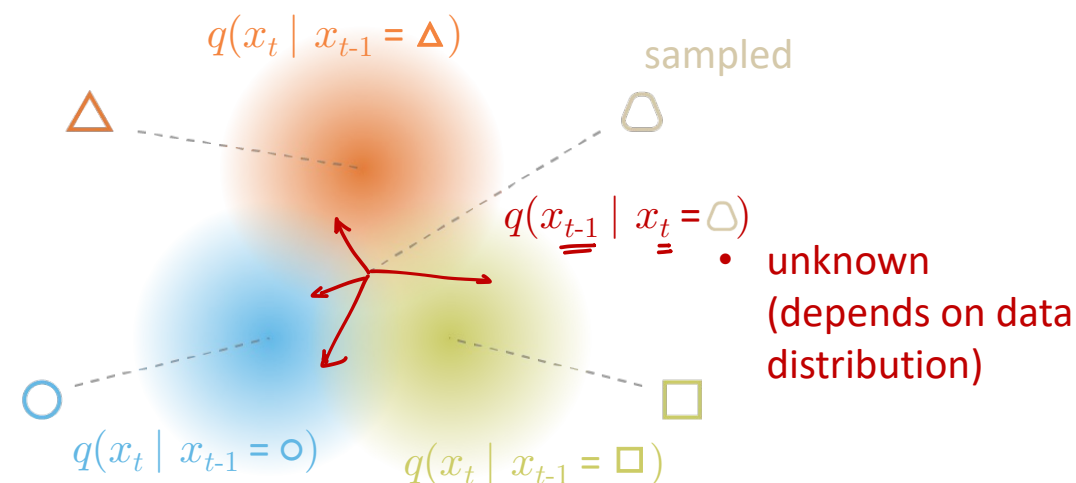
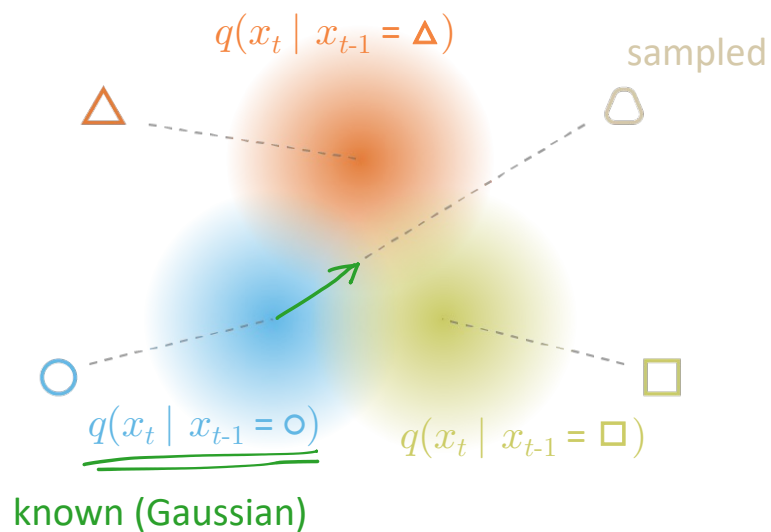
- Recreate true samples from a Gaussian noise if we can sample from $q(\mathbf{x}_{t-1}|\mathbf{x}_t)$
- Learn to approximate $q(\mathbf{x}_{t-1}|\mathbf{x}_t)$ with $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$

$$p_\theta(\mathbf{x}_{0:T}) = p(\mathbf{x}_T) \prod_{t=1}^T p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) \quad p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_\theta(\mathbf{x}_t, t), \boldsymbol{\Sigma}_\theta(\mathbf{x}_t, t))$$



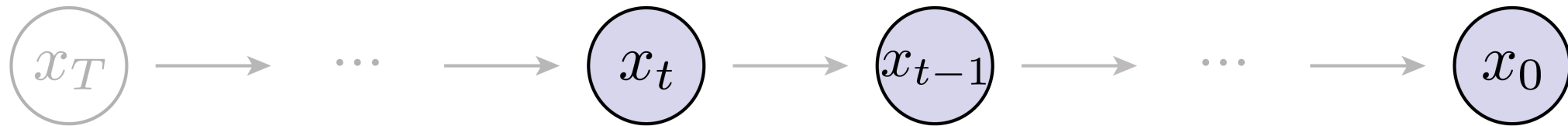
Reverse Process

- Recreate true samples from a Gaussian noise if we can sample from $q(\mathbf{x}_{t-1}|\mathbf{x}_t)$
- Learn to approximate $q(\mathbf{x}_{t-1}|\mathbf{x}_t)$ with $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$
- **Why $q(\mathbf{x}_{t-1}|\mathbf{x}_t)$ is unknown?**



Reverse Process

- Approximate $q(\mathbf{x}_{t-1}|\mathbf{x}_t)$ with $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$
- From $q(\mathbf{x}_{t-1}|\mathbf{x}_t)$ to $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$



$$q(x_{t-1} \mid x_t, x_0)$$

$$= \mathcal{N}(x_{t-1} \mid \tilde{\mu}(x_t, x_0), \tilde{\beta}_t \mathbf{I})$$

Reverse Process

- From $q(\mathbf{x}_{t-1}|\mathbf{x}_t)$ to $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$

$$q(x_{t-1} | x_t, x_0) = \mathcal{N}(x_{t-1} | \tilde{\mu}(x_t, x_0), \tilde{\beta}_t \mathbf{I})$$

$$\tilde{\mu}_t(x_t, x_0) := \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t}x_0 + \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}x_t$$

mean

$$\tilde{\beta}_t = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t}\beta_t$$

var

$$\begin{aligned} q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) &= q(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{x}_0) \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)} \\ &\propto \exp\left(-\frac{1}{2}\left(\frac{(\mathbf{x}_t - \sqrt{\alpha_t}\mathbf{x}_{t-1})^2}{\beta_t} + \frac{(\mathbf{x}_{t-1} - \sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_0)^2}{1 - \bar{\alpha}_{t-1}} - \frac{(\mathbf{x}_t - \sqrt{\bar{\alpha}_t}\mathbf{x}_0)^2}{1 - \bar{\alpha}_t}\right)\right) \\ &= \exp\left(-\frac{1}{2}\left(\frac{\mathbf{x}_t^2 - 2\sqrt{\alpha_t}\mathbf{x}_t\mathbf{x}_{t-1} + \alpha_t\mathbf{x}_{t-1}^2}{\beta_t} + \frac{\mathbf{x}_{t-1}^2 - 2\sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_0\mathbf{x}_{t-1} + \bar{\alpha}_{t-1}\mathbf{x}_0^2}{1 - \bar{\alpha}_{t-1}} - \frac{(\mathbf{x}_t - \sqrt{\bar{\alpha}_t}\mathbf{x}_0)^2}{1 - \bar{\alpha}_t}\right)\right) \\ &= \exp\left(-\frac{1}{2}\left(\left(\frac{\alpha_t}{\beta_t} + \frac{1}{1 - \bar{\alpha}_{t-1}}\right)\mathbf{x}_{t-1}^2 - \left(\frac{2\sqrt{\alpha_t}}{\beta_t}\mathbf{x}_t + \frac{2\sqrt{\bar{\alpha}_{t-1}}}{1 - \bar{\alpha}_{t-1}}\mathbf{x}_0\right)\mathbf{x}_{t-1} + C(\mathbf{x}_t, \mathbf{x}_0)\right)\right) \end{aligned}$$

Reverse Process

- From $q(\mathbf{x}_{t-1}|\mathbf{x}_t)$ to $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$

$$q(x_{t-1} | x_t, x_0) = \mathcal{N}(x_{t-1} | \tilde{\mu}(x_t, x_0), \tilde{\beta}_t \mathbf{I})$$

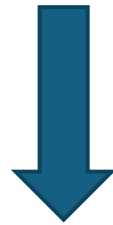
$$\tilde{\mu}_t(x_t, x_0) := \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t}x_0 + \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}x_t$$

mean

$$\tilde{\beta}_t = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t}\beta_t$$

var

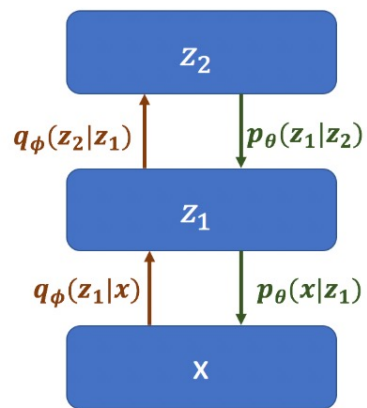
$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon$$



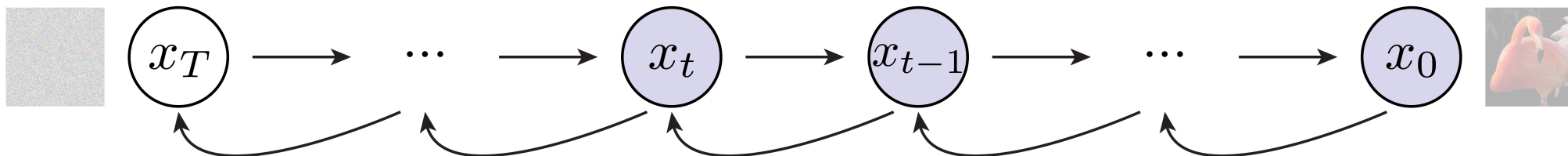
$$= \frac{1}{\sqrt{\bar{\alpha}_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon \right) \text{ noise}$$

Learning Objective

- Recall hierarchical VAE

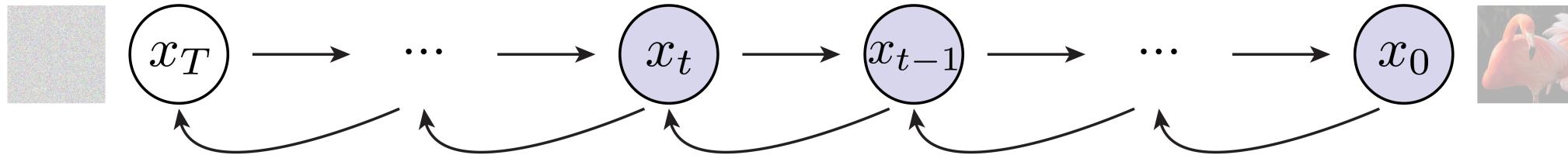


Hierarchical VAE



Learning Objective

- Recall hierarchical VAE



$$\begin{aligned}
 -\log p_\theta(\mathbf{x}_0) &\leq -\log p_\theta(\mathbf{x}_0) + D_{\text{KL}}(q(\mathbf{x}_{1:T}|\mathbf{x}_0)||p_\theta(\mathbf{x}_{1:T}|\mathbf{x}_0)) && \text{; KL is non-negative} \\
 &= -\log p_\theta(\mathbf{x}_0) + \mathbb{E}_{\mathbf{x}_{1:T} \sim q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[\log \frac{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}{p_\theta(\mathbf{x}_{0:T})/p_\theta(\mathbf{x}_0)} \right] \\
 &= -\log p_\theta(\mathbf{x}_0) + \mathbb{E}_q \left[\log \frac{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}{p_\theta(\mathbf{x}_{0:T})} + \log p_\theta(\mathbf{x}_0) \right] \\
 &= \mathbb{E}_q \left[\log \frac{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}{p_\theta(\mathbf{x}_{0:T})} \right]
 \end{aligned}$$

$$\text{Let } L_{\text{VLB}} = \mathbb{E}_{q(\mathbf{x}_{0:T})} \left[\log \frac{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}{p_\theta(\mathbf{x}_{0:T})} \right] \geq -\mathbb{E}_{q(\mathbf{x}_0)} \log p_\theta(\mathbf{x}_0)$$

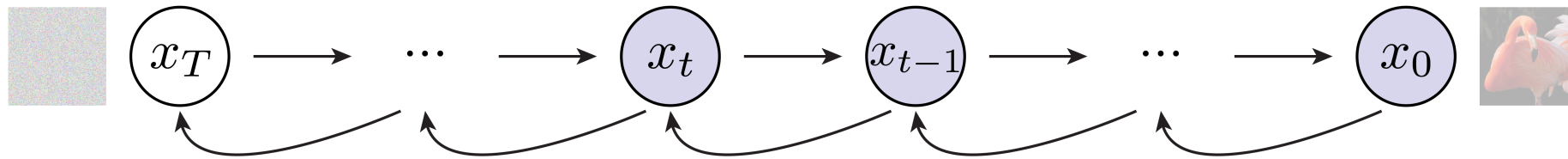
VAE's objective:

$$\mathcal{L}_{\theta, \phi}(x) = -\mathbb{E}_{z \sim q_\phi(z|x)} \left[\log p_\theta(x|z) \right] + \mathcal{D}_{\text{KL}}(q_\phi(z|x)||p(z))$$

$$\begin{aligned}
 L_{\text{VLB}} &= \mathbb{E}_{q(\mathbf{x}_{0:T})} \left[\log \frac{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}{p_\theta(\mathbf{x}_{0:T})} \right] \\
 &= \mathbb{E}_q \left[\log \frac{\prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1})}{p_\theta(\mathbf{x}_T) \prod_{t=1}^T p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)} \right] \\
 &= \mathbb{E}_q \left[-\log p_\theta(\mathbf{x}_T) + \sum_{t=1}^T \log \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1})}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)} \right] \\
 &= \mathbb{E}_q \left[-\log p_\theta(\mathbf{x}_T) + \sum_{t=2}^T \log \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1})}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)} + \log \frac{q(\mathbf{x}_1|\mathbf{x}_0)}{p_\theta(\mathbf{x}_0|\mathbf{x}_1)} \right] \\
 &= \mathbb{E}_q \left[-\log p_\theta(\mathbf{x}_T) + \sum_{t=2}^T \log \left(\frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)} \cdot \frac{q(\mathbf{x}_t|\mathbf{x}_0)}{q(\mathbf{x}_{t-1}|\mathbf{x}_0)} \right) + \log \frac{q(\mathbf{x}_1|\mathbf{x}_0)}{p_\theta(\mathbf{x}_0|\mathbf{x}_1)} \right] \\
 &= \mathbb{E}_q \left[-\log p_\theta(\mathbf{x}_T) + \sum_{t=2}^T \log \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)} + \sum_{t=2}^T \log \frac{q(\mathbf{x}_t|\mathbf{x}_0)}{q(\mathbf{x}_{t-1}|\mathbf{x}_0)} + \log \frac{q(\mathbf{x}_1|\mathbf{x}_0)}{p_\theta(\mathbf{x}_0|\mathbf{x}_1)} \right] \\
 &= \mathbb{E}_q \left[-\log p_\theta(\mathbf{x}_T) + \sum_{t=2}^T \log \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)} + \log \frac{q(\mathbf{x}_T|\mathbf{x}_0)}{q(\mathbf{x}_1|\mathbf{x}_0)} + \log \frac{q(\mathbf{x}_1|\mathbf{x}_0)}{p_\theta(\mathbf{x}_0|\mathbf{x}_1)} \right] \\
 &= \mathbb{E}_q \left[\log \frac{q(\mathbf{x}_T|\mathbf{x}_0)}{p_\theta(\mathbf{x}_T)} + \sum_{t=2}^T \log \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)} - \log p_\theta(\mathbf{x}_0|\mathbf{x}_1) \right] \\
 &= \mathbb{E}_q \left[\underbrace{D_{\text{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p_\theta(\mathbf{x}_T))}_{L_T} + \sum_{t=2}^T \underbrace{D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{L_{t-1}} - \log p_\theta(\mathbf{x}_0|\mathbf{x}_1) \right]
 \end{aligned}$$

Learning Objective

- Recall hierarchical VAE



$$\mathcal{L}_{\text{VLB}} := \mathcal{L}_T + \mathcal{L}_{T-1} + \dots + \mathcal{L}_0$$

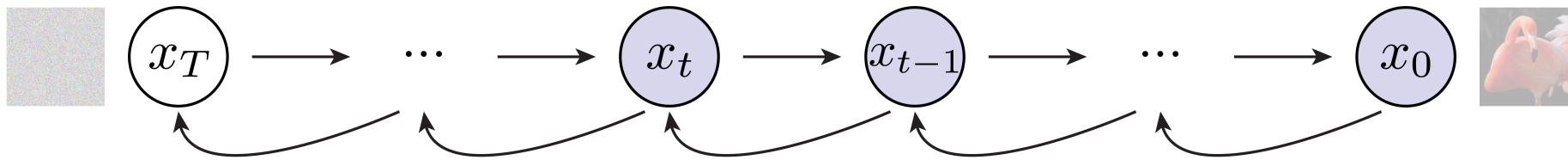
$$\mathcal{L}_T := D_{\text{KL}}\left(q(x_T \mid x_0) \parallel p_{\theta}(x_T)\right)$$

$$\mathcal{L}_{t-1} := D_{\text{KL}}\left(q(x_{t-1} \mid x_t, x_0) \parallel p_{\theta}(x_{t-1} \mid x_t)\right)$$

$$\mathcal{L}_0 := -\log p_{\theta}(x_0 \mid x_1)$$

Learning Objective

- Recall hierarchical VAE



$$\mathcal{L}_{\text{VLB}} := \mathcal{L}_T + \mathcal{L}_{T-1} + \dots + \mathcal{L}_0$$

no parameter, unlike VAE's q_ϕ

$$\text{constant } \mathcal{L}_T := D_{\text{KL}} \left(\underline{q(x_T | x_0)} \parallel \underline{p_\theta(x_T)} \right)$$

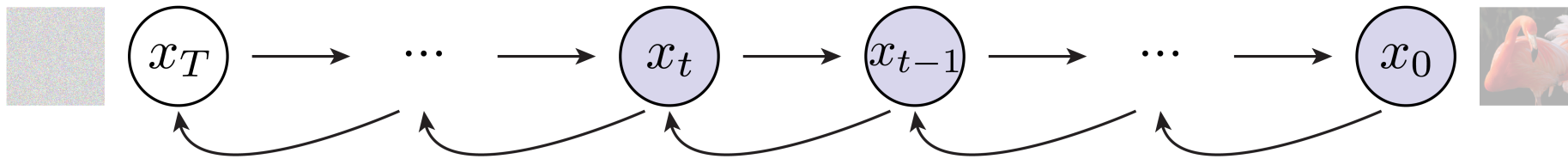
Gaussian

$$\mathcal{L}_{t-1} := D_{\text{KL}} \left(q(x_{t-1} | x_t, x_0) \parallel p_\theta(x_{t-1} | x_t) \right)$$

$$\mathcal{L}_0 := -\log p_\theta(x_0 | x_1)$$

Learning Objective

- Recall hierarchical VAE



$$\mathcal{L}_{\text{VLB}} := \mathcal{L}_T + \mathcal{L}_{T-1} + \dots + \mathcal{L}_0$$

$$\mathcal{L}_T := D_{\text{KL}}\left(q(x_T \mid x_0) \parallel p_{\theta}(x_T)\right)$$

$$\mathcal{L}_{t-1} := D_{\text{KL}}\left(q(x_{t-1} \mid x_t, x_0) \parallel p_{\theta}(x_{t-1} \mid x_t)\right)$$

reconstruction loss,
like VAE

$$\mathcal{L}_0 := -\log p_{\theta}(x_0 \mid x_1)$$

Learning Objective

- Recall hierarchical VAE

$$\mathcal{L}_{t-1} := D_{\text{KL}}\left(q(x_{t-1} \mid x_t, x_0) \parallel p_{\theta}(x_{t-1} \mid x_t)\right)$$



$$q(x_{t-1} \mid x_t, x_0) = \mathcal{N}(x_{t-1} \mid \tilde{\mu}(x_t, x_0), \tilde{\beta}_t \mathbf{I})$$

$$\tilde{\beta}_t := \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \beta_t$$

$$\tilde{\mu}_t(x_t, x_0) := \frac{1}{\sqrt{\bar{\alpha}_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon \right)$$

$$p_{\theta}(\mathbf{x}_{t-1} \mid \mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_{\theta}(\mathbf{x}_t, t), \boldsymbol{\Sigma}_{\theta}(\mathbf{x}_t, t))$$

$$p_{\theta}(\mathbf{x}_{t-1} \mid \mathbf{x}_t) = \mathcal{N}\left(\mathbf{x}_{t-1}; \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \boldsymbol{\epsilon}_{\theta}(\mathbf{x}_t, t) \right), \boldsymbol{\Sigma}_{\theta}(\mathbf{x}_t, t)\right)$$

$$\boldsymbol{\Sigma}_{\theta}(\mathbf{x}_t, t) = \sigma_t^2 \mathbf{I}$$

Learning Objective

- Recall hierarchical VAE

$$\begin{aligned}
 \mathcal{L}_{t-1} &:= D_{\text{KL}}\left(q(x_{t-1} \mid x_t, x_0) \parallel p_{\theta}(x_{t-1} \mid x_t)\right) \\
 &= \mathbb{E}_{\mathbf{x}_0, \epsilon} \left[\frac{1}{2 \|\Sigma_{\theta}(\mathbf{x}_t, t)\|_2^2} \|\tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0) - \mu_{\theta}(\mathbf{x}_t, t)\|^2 \right] \\
 &= \mathbb{E}_{\mathbf{x}_0, \epsilon} \left[\frac{1}{2 \|\Sigma_{\theta}\|_2^2} \left\| \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_t \right) - \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_{\theta}(\mathbf{x}_t, t) \right) \right\|^2 \right] \\
 &= \mathbb{E}_{\mathbf{x}_0, \epsilon} \left[\frac{(1 - \alpha_t)^2}{2 \alpha_t (1 - \bar{\alpha}_t) \|\Sigma_{\theta}\|_2^2} \|\epsilon_t - \epsilon_{\theta}(\mathbf{x}_t, t)\|^2 \right] \\
 &= \mathbb{E}_{\mathbf{x}_0, \epsilon} \left[\frac{(1 - \alpha_t)^2}{2 \alpha_t (1 - \bar{\alpha}_t) \|\Sigma_{\theta}\|_2^2} \|\epsilon_t - \epsilon_{\theta}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon_t, t)\|^2 \right]
 \end{aligned}$$

Learning Objective

- Recall hierarchical VAE

$$\mathbb{E}_{\mathbf{x}_0, \epsilon} \left[\frac{(1 - \alpha_t)^2}{2\alpha_t(1 - \bar{\alpha}_t) \|\Sigma_\theta\|_2^2} \|\epsilon_t - \epsilon_\theta(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon_t, t)\|^2 \right]$$

$$\mathcal{L} = \mathbb{E}_{x_0, t, \epsilon} \left[w_t \|\epsilon - \epsilon_\theta(x_t, t)\|^2 \right]$$

Diagram illustrating the expectation over different variables in the loss function $\mathcal{L}$:

- Red arrow: x_0 is sampled over p_{data} .
- Green arrow: t is sampled over $[1, T]$.
- Orange arrow: ϵ is sampled over $\mathcal{N}(0, \mathbf{I})$.

Learning Objective

- Recall hierarchical VAE

$$\mathcal{L} = \mathbb{E}_{x_0, t, \epsilon} \left[w_t \| \epsilon - \epsilon_\theta(x_t, t) \|^2 \right]$$

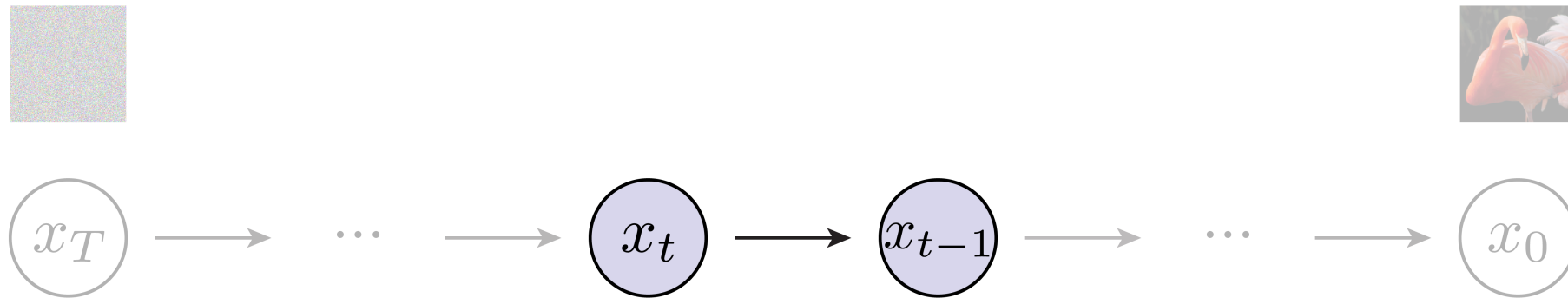
Diagram illustrating the expectation over parameters:

- Red arrow: over p_{data} (points to x_0)
- Green arrow: over $[1, T]$ (points to t)
- Orange arrow: over $\mathcal{N}(0, \mathbf{I})$ (points to ϵ)

Objective	IS	FID
L , learned diagonal Σ	–	–
L , fixed isotropic Σ	7.67 ± 0.13	13.51
$\ \tilde{\epsilon} - \epsilon_\theta\ ^2$ (L_{simple})	9.46 ± 0.11	3.17

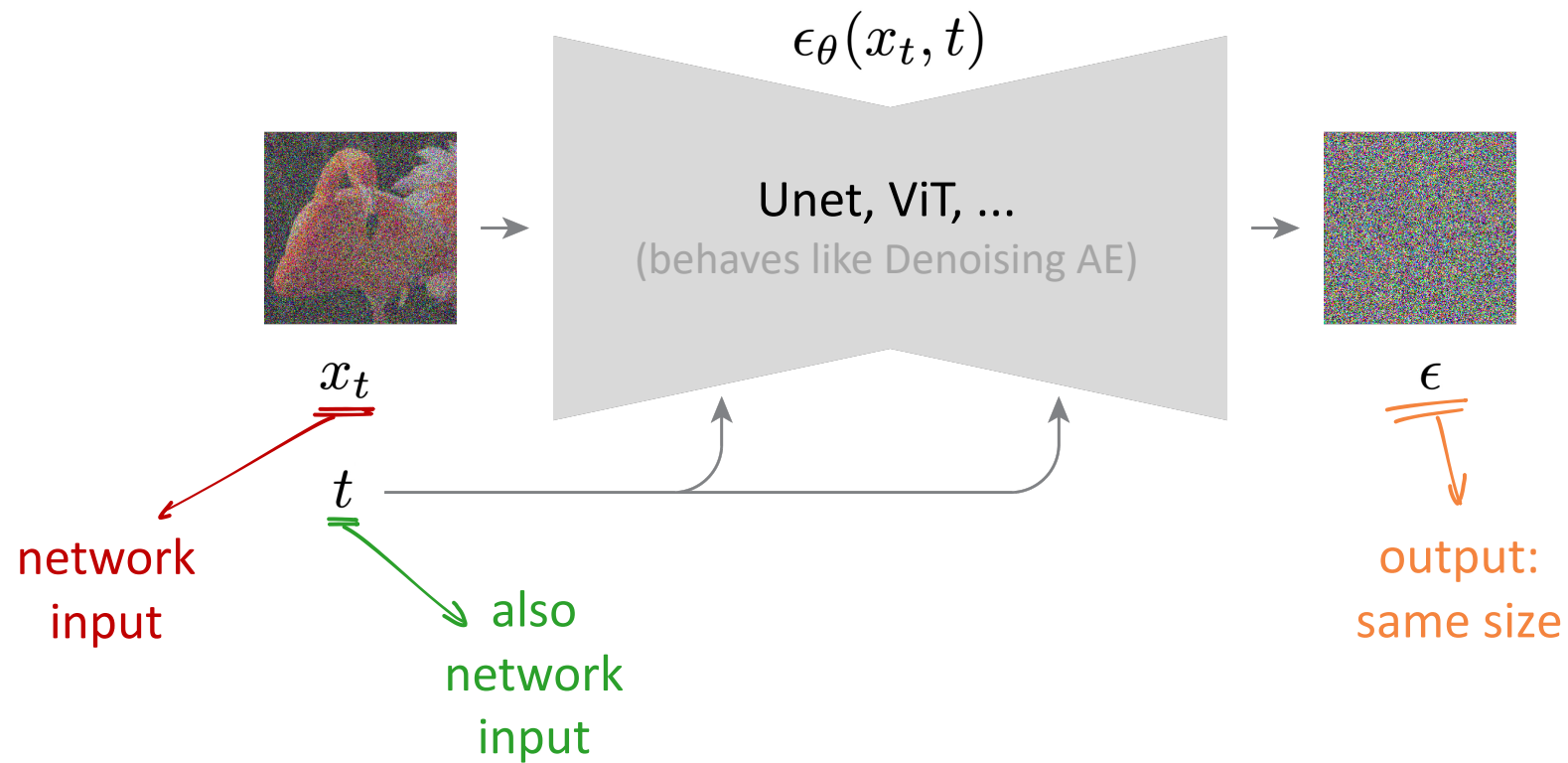
Representing $p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)$ with neural networks

- **Noise Conditional Network**
 - Diffusion models decompose a distribution into many simpler ones.
 - We need the same # networks to fit all of them.
 - We can combine all into one “powerful” network.
 - This network is conditioned on noise level t .



Representing $p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)$ with neural networks

- Noise Conditional Network



Overall

- Simple and effective

Algorithm 1 Training

```

1: repeat
2:    $\mathbf{x}_0 \sim q(\mathbf{x}_0)$ 
3:    $t \sim \text{Uniform}(\{1, \dots, T\})$ 
4:    $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
5:   Take gradient descent step on
        $\nabla_{\theta} \|\epsilon - \epsilon_{\theta}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t)\|^2$ 
6: until converged
    
```

x_t

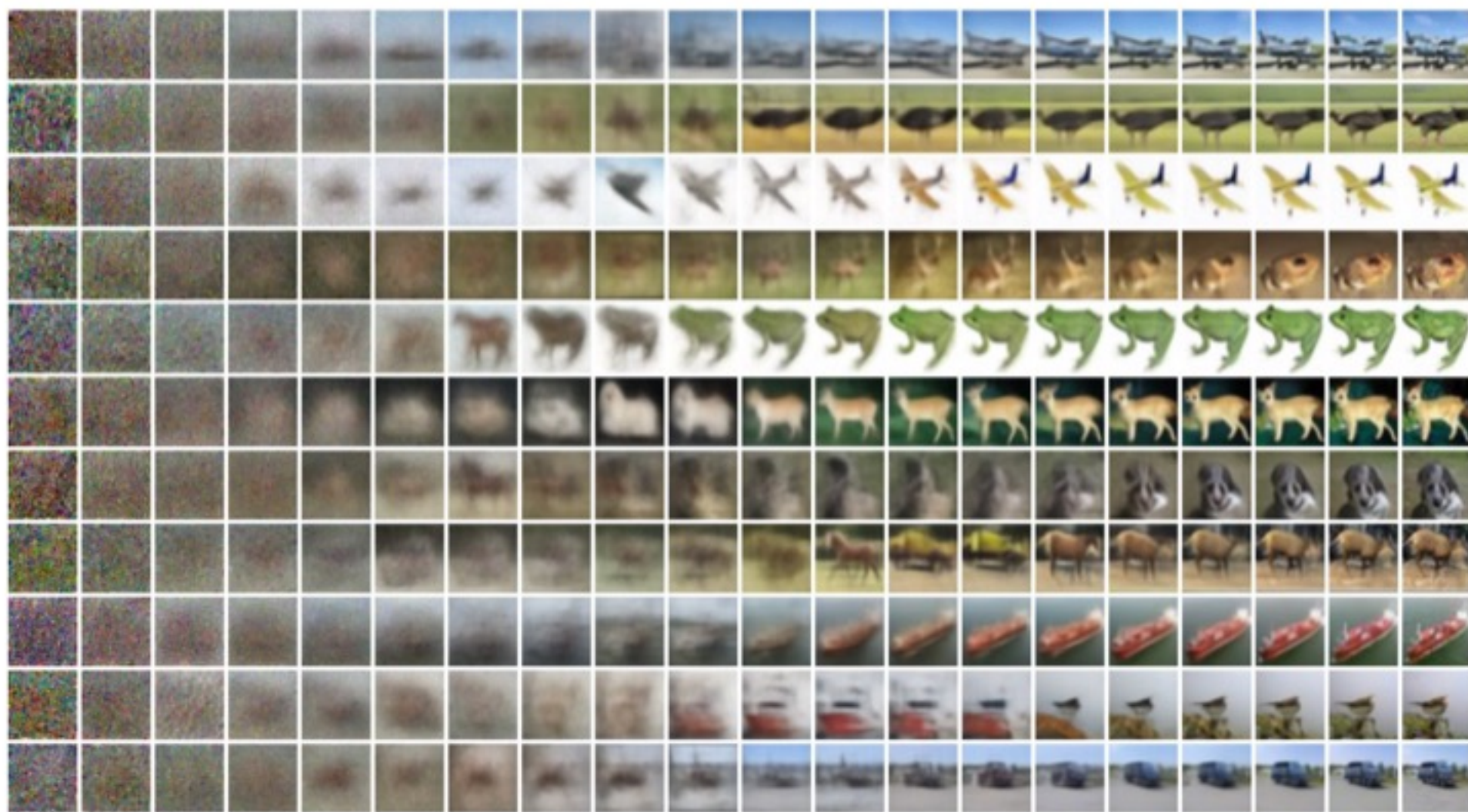
Algorithm 2 Sampling

```

1:  $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
2: for  $t = T, \dots, 1$  do
3:    $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$  if  $t > 1$ , else  $\mathbf{z} = \mathbf{0}$ 
4:    $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left( \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_{\theta}(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$ 
5: end for
6: return  $\mathbf{x}_0$ 
    
```

estimated μ

sampling from
estimated distribution



Shared intermediate latents



Already viewed diffusion models in the perspective of **hierarchical VAE**

- Other perspectives
 - **Score matching**

Score matching is one solution to **Energy-based models**

Probability distributions $p(x)$ are a key building block in generative modeling

- Non-negative: $p(x) \geq 0$
- Sum-to-one

Finding a non-negative function is not hard

Sum-to-one is key challenge

- choose $p_\theta(x)$ so that we know the sum **analytically** as a function of θ

Score matching is one solution to Energy-based models

- Energy-based models
 - Relax the assumption: choose $p_{\theta}(\mathbf{x})$ so that we know the sum **analytically** as a function of θ

$$p_{\theta}(\mathbf{x}) = \frac{1}{\int \exp(f_{\theta}(\mathbf{x})) d\mathbf{x}} \exp(f_{\theta}(\mathbf{x})) = \frac{1}{Z(\theta)} \exp(f_{\theta}(\mathbf{x}))$$

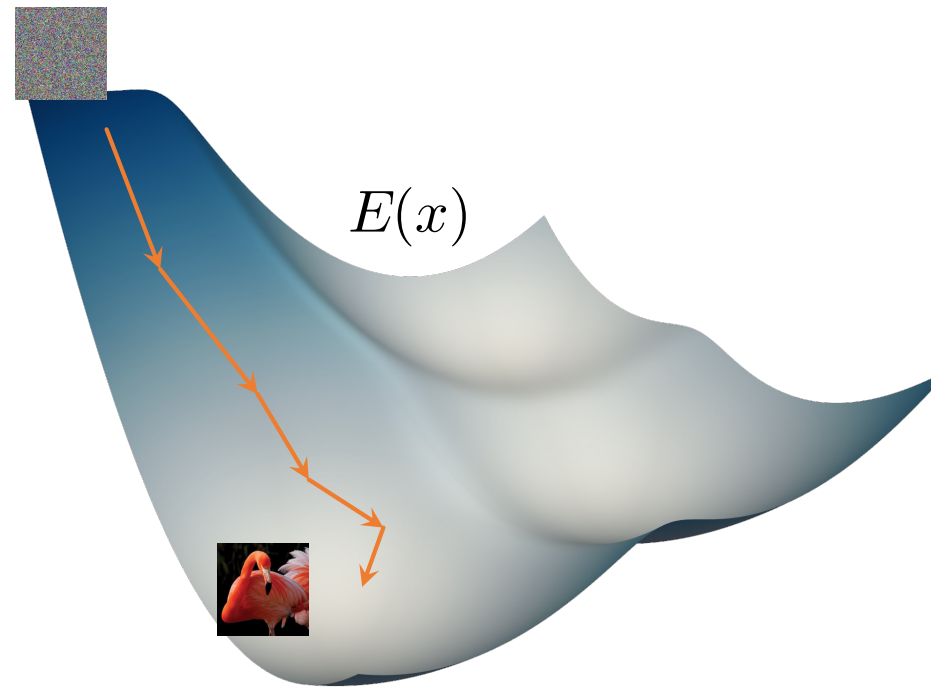
- The normalization constant (the partition function)

$$Z(\theta) = \int \exp(f_{\theta}(\mathbf{x})) d\mathbf{x}$$

- $-f_{\theta}(\mathbf{x})$ is called **the energy**, hence the name.
- Intuitively, **configurations $\mathbf{x}$ with low energy (high $f_{\theta}(\mathbf{x})$) are more likely.**
- **Computing $Z(\theta)$ is infeasible but some tasks do not require knowing $Z(\theta)$.**

Score matching is one solution to Energy-based models

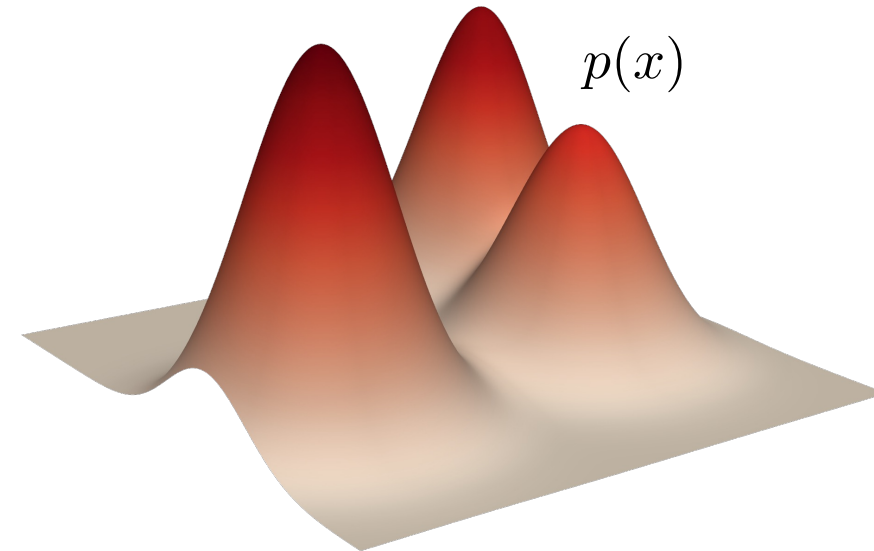
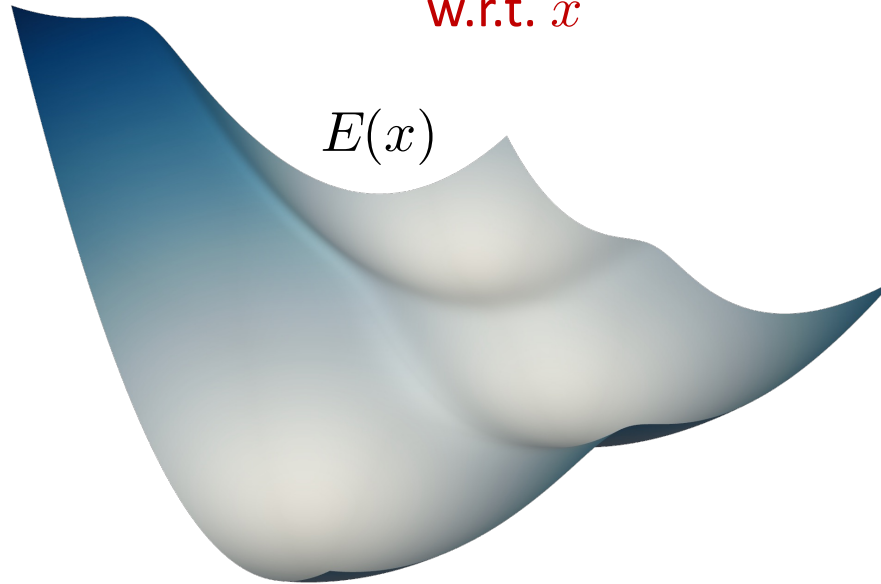
- Energy-based models



Score matching is one solution to Energy-based models

- Energy-based models
 - **Score function: gradient of log-probability**

$$\underbrace{\nabla_x}_{\text{w.r.t. } x} \log p(x) = -\nabla_x E(x) - \underbrace{\nabla_x \log Z}_{=0}$$

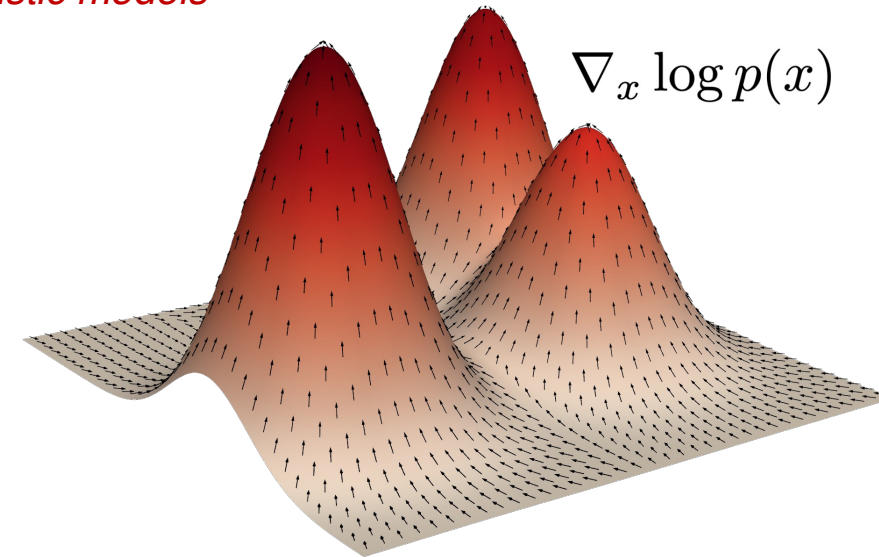
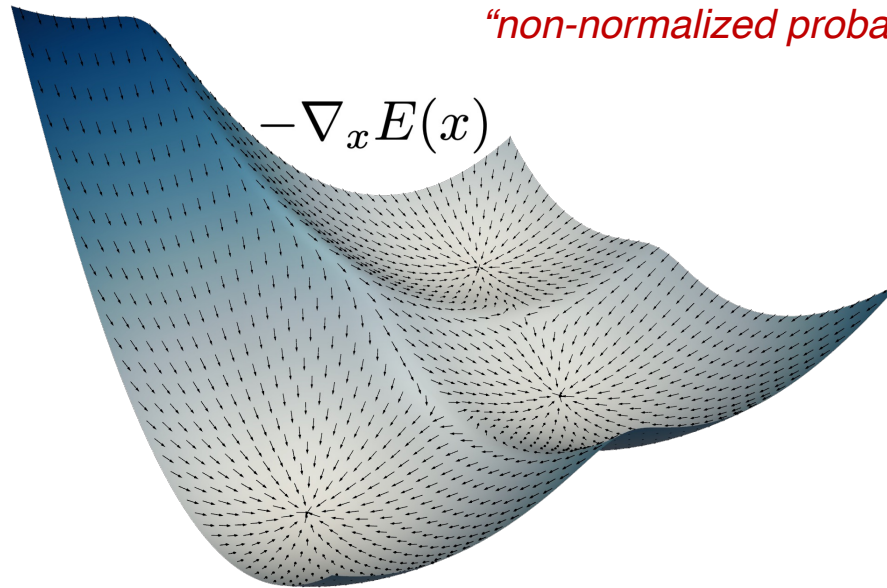


Score matching is one solution to Energy-based models

- Energy-based models
 - **Score function: gradient of log-probability**

$$\nabla_x \log p(x) = -\nabla_x E(x)$$

“non-normalized probabilistic models”

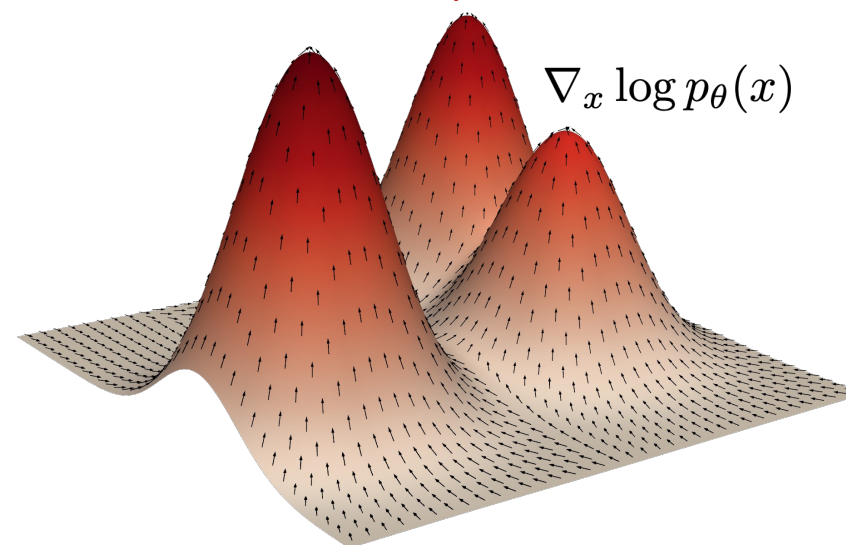
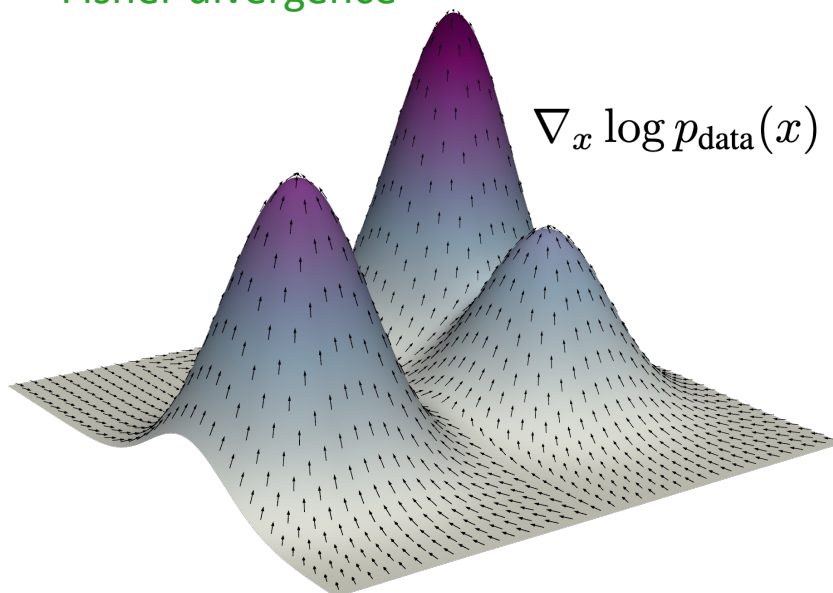


*only visualize directions

Score matching

- Instead of parameterizing p , we can parameterize the score

$$\underbrace{D_F(p_{\text{data}}(x) \parallel p_{\theta}(x))}_{\text{Fisher divergence}} = \mathbb{E}_{p_{\text{data}}(x)} \left[\frac{1}{2} \left\| \underbrace{\nabla_x \log p_{\text{data}}(x)}_{\text{score of data}} - \underbrace{\nabla_x \log p_{\theta}(x)}_{\text{parameterized score}} \right\|^2 \right]$$

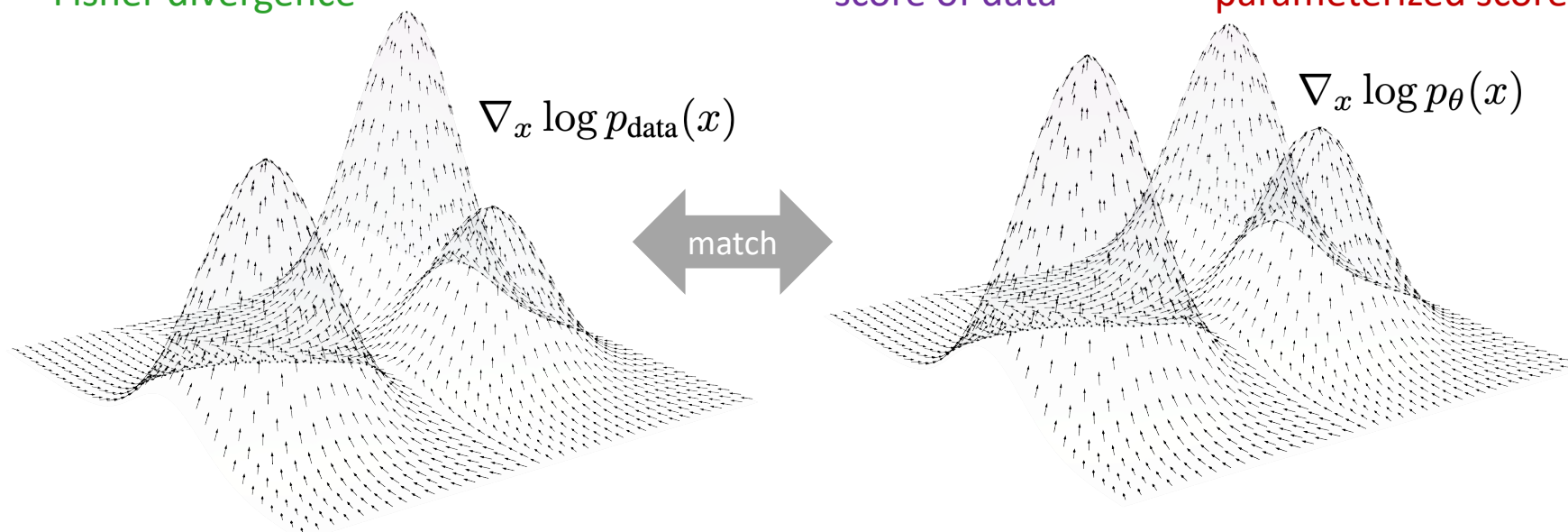


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*only visualize directions

Score matching

- Instead of parameterizing p , we can parameterize the score
- Denoising score matching
 - With noised data $\tilde{x} := x + \epsilon$

$$\underbrace{D_F(q(\tilde{x}) \parallel p_\theta(\tilde{x}))}_{\text{Fisher divergence of noised data}} = \underbrace{\mathbb{E}_{q(x, \tilde{x})}}_{\text{joint distribution}} \left[\frac{1}{2} \left\| \underbrace{\nabla_{\tilde{x}} \log q(\tilde{x} \mid x)}_{\text{score of conditional}} - \underbrace{\nabla_{\tilde{x}} \log p_\theta(\tilde{x})}_{\text{parameterized score}} \right\|^2 \right] + \text{constant}$$

Score matching

- Given a score function, we can sample x from p by iterating

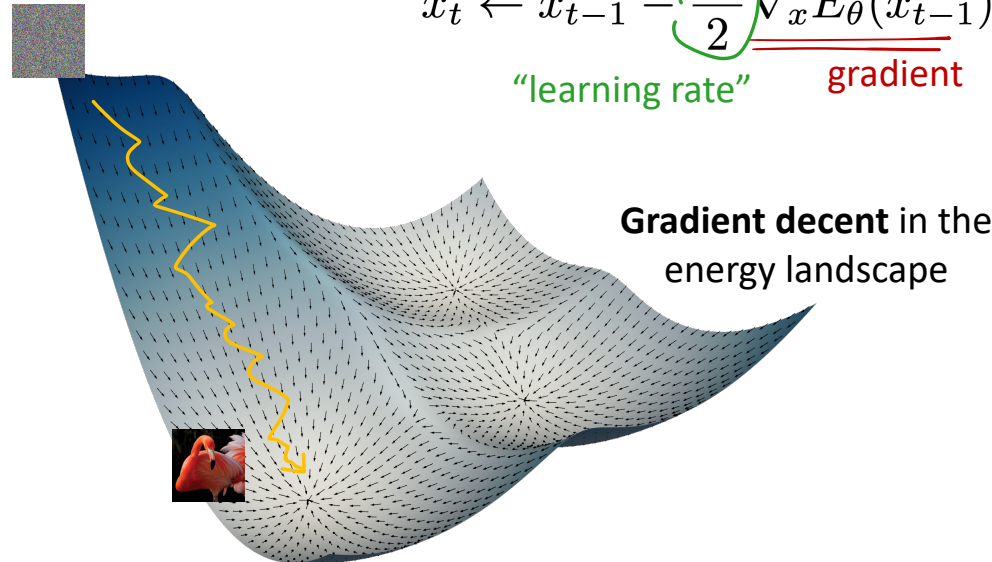
$$x_t \leftarrow x_{t-1} + \underbrace{\left(\frac{\sigma^2}{2}\right)}_{\text{step size}} \underbrace{\nabla_x \log p_\theta(x_{t-1})}_{\substack{\text{score function} \\ \text{(don't need to know } p)}} + \underbrace{\sigma z_t}_{\mathcal{N}(0, \mathbf{I})}$$

$\swarrow$ (neg) gradient of energy
 $-\nabla_x E_\theta(x_{t-1})$

Score matching

- Given a score function, we can sample x from p by iterating

$$x_t \leftarrow x_{t-1} - \underbrace{\frac{\sigma^2}{2}}_{\text{"learning rate"}} \underbrace{\nabla_x E_\theta(x_{t-1})}_{\text{gradient}} + \underbrace{\sigma z_t}_{\text{perturbation}}$$



Score matching

- Given a score function, we can sample x from p by iterating

$$x_t \leftarrow x_{t-1} - \underbrace{\frac{\sigma^2}{2}}_{\text{"learning rate"}} \underbrace{\nabla_x E_\theta(x_{t-1})}_{\text{gradient}} + \underbrace{\sigma z_t}_{\text{perturbation}}$$

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6: until converged
    
```

score function

Algorithm 2 Sampling

```

1:  $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
2: for  $t = T, \dots, 1$  do
3:    $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$  if  $t > 1$ , else  $\mathbf{z} = \mathbf{0}$ 
4:    $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left( \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$ 
5: end for
6: return  $\mathbf{x}_0$ 
    
```

score function

Langevin Dynamics



To be continued